WORKED SOLUTION

STEP III 1992, Question 16

Probability and Statistics • Poisson Thinning

STEP III 1992, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q16

The key idea

Once the total number of misprints is fixed, each one is an independent trial that I either spot or miss, so part (i) is a binomial probability. Part (ii) is a conditional probability, and the denominator, the chance that I spot exactly m misprints, needs us to add up over every possible number of missed misprints. That sum turns out to be an exponential series, which is what makes everything collapse to a neat Poisson form.

Solution

Throughout, write $q = 1 - p$. Let N be the total number of misprints in the issue, S the number I spot and $U = N - S$ the number I fail to spot. We are told that

$$\mathrm{P}(N = k) = \frac{\mathrm{e}^{-\lambda}\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

Part (i)

Given $N = m + n$, each of the $m + n$ misprints is spotted with probability p, independently of the others. So, conditional on $N = m + n$, the number spotted has the binomial distribution $\mathrm{B}(m + n, p)$. Any particular choice of m spotted and n missed misprints has probability $p^m q^n$, and there are $\binom{m+n}{m}$ such choices, so

$$\mathrm{P}(S = m \mid N = m + n) = \binom{m+n}{m} p^m (1-p)^n.$$

Part (ii)

We want $\mathrm{P}(U = n \mid S = m)$. The event $\{S = m,\ U = n\}$ is the same as $\{N = m+n,\ S = m\}$, so by part (i)

$$\begin{aligned}\mathrm{P}(S = m,\ U = n) &= \mathrm{P}(N = m+n)\,\mathrm{P}(S = m \mid N = m+n)\\ &= \frac{\mathrm{e}^{-\lambda}\lambda^{m+n}}{(m+n)!}\cdot\frac{(m+n)!}{m!\,n!}\,p^m q^n\\ &= \mathrm{e}^{-\lambda}\,\frac{(\lambda p)^m}{m!}\cdot\frac{(\lambda q)^n}{n!}.\end{aligned}$$

The factor $(m+n)!$ cancels, which is what makes this question work.

> **Ask yourself** How do we find $\mathrm{P}(S = m)$, when we don't know how many misprints there were?

If I spot exactly m, the number I missed could be any of $0, 1, 2, \ldots$, and these cases are mutually exclusive. So we sum the joint probability over all values j of U:

$$\begin{aligned}\mathrm{P}(S = m) &= \sum_{j=0}^{\infty}\mathrm{e}^{-\lambda}\,\frac{(\lambda p)^m}{m!}\cdot\frac{(\lambda q)^j}{j!} = \mathrm{e}^{-\lambda}\,\frac{(\lambda p)^m}{m!}\,\mathrm{e}^{\lambda q}\\ &= \frac{(\lambda p)^m}{m!}\,\mathrm{e}^{-\lambda p},\end{aligned}$$

using $\sum_{j\geq 0} x^j/j! = \mathrm{e}^x$ and $-\lambda + \lambda q = -\lambda p$. (So the number of misprints I spot is itself Poisson, with mean λp.) This is positive, so we may divide by it. Therefore

$$\begin{aligned}\mathrm{P}(U = n \mid S = m) &= \frac{\mathrm{P}(S = m,\ U = n)}{\mathrm{P}(S = m)} = \frac{\mathrm{e}^{-\lambda}\,\dfrac{(\lambda p)^m}{m!}\cdot\dfrac{(\lambda q)^n}{n!}}{\mathrm{e}^{-\lambda p}\,\dfrac{(\lambda p)^m}{m!}}\\ &= \frac{(\lambda q)^n}{n!}\,\mathrm{e}^{-\lambda+\lambda p} = \frac{(1-p)^n\lambda^n}{n!}\,\mathrm{e}^{-(1-p)\lambda},\end{aligned}$$

as required.

The answer doesn't depend on m at all: knowing how many misprints I spotted tells me nothing about how many I missed. The number missed is Poisson with mean $(1-p)\lambda$ whatever m is.

Checks

- Summing the final answer over $n = 0, 1, 2, \ldots$ gives $\mathrm{e}^{(1-p)\lambda}\mathrm{e}^{-(1-p)\lambda} = 1$, as a conditional distribution must.
- As $p \to 1$ I spot everything, and the answer tends to 1 for $n = 0$ and 0 for $n \geq 1$. Good.

- With $\lambda = 3$, $p = 0.6$, $m = 2$, $n = 1$ the formula gives $1.2\,\mathrm{e}^{-1.2} \approx 0.3614$. A simulation of 400 000 newspapers, keeping those in which exactly two misprints were spotted, gave 0.360.

Common slips

- Forgetting the binomial coefficient $\binom{m+n}{m}$ in part (i): the m spotted misprints can be any m of the $m + n$.
- Taking $\mathrm{P}(S = m)$ to be $\mathrm{P}(N = m)$, or to be the part (i) answer, rather than summing over all possible numbers of missed misprints.

Beyond STEP

This is the *thinning* (or splitting) property of the Poisson distribution: if a Poisson(λ) number of objects are each kept with probability p, the kept and discarded counts are *independent* Poisson variables with means λp and $\lambda(1 - p)$. The joint probability we found factorises into exactly these two pieces. Thinning is a key tool in the theory of Poisson processes, where it lets you split one stream of random events into independent sub-streams. Grimmett and Welsh, *Probability: An Introduction*, covers it, along with the generating-function methods that prove it in one line.

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